

Nonparametric Predictive Regression with Exogenous Shocks: A Sieve Empirical Likelihood Approach (In progress)

Siyan Dong

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1 Introduction and Data-Generating Process

We consider the problem of testing return predictability in the presence of highly persistent predictors and stationary exogenous covariates. The exogenous covariate may be a scalar or d -dimensional vector of macroeconomic, environmental, or other predetermined state variables whose effect on returns is not well described by a constant intercept. The data-generating process (DGP) is specified by the following system of equations:

1. Predictive Regression:

$$Y_t = m(W_{t-1}) + \beta X_{t-1} + U_t \quad (1)$$

$$X_t = \theta + \rho X_{t-1} + \varepsilon_t \quad (2)$$

2. Endogeneity:

$$U_t = \phi \varepsilon_t + z_t \quad \text{with } E[z_t | \varepsilon_t] = 0 \quad (3)$$

where Y_t represents asset returns, $W_{t-1} \in \mathbb{R}^d$ is a stationary exogenous covariate vector, and X_{t-1} is a persistent financial predictor. The function $m(W_{t-1})$ is an unknown nonparametric nuisance component: it can be viewed as a flexible intercept or exogenous-control term that absorbs the direct effect of W_{t-1} on returns. The key timing restriction is contemporaneous sieve orthogonality between the exogenous covariate vector and the persistent predictor weight at the sampling frequency used for inference. This restriction does not rule out effects of W_{t-1} on returns through $m(W_{t-1})$, nor does it rule out adjustment in future values of X_t .

Throughout the theoretical development, W_{t-1} should be read as a generic stationary exogenous covariate vector and is not restricted to be univariate; the scalar case corresponds to $d = 1$. The weather-shock interpretation used below is one empirical application of this more general setup.

Motivated by Perron (1989), our goal is to investigate return predictability while avoiding size distortions or false inferences caused by near unit roots in the predictor dynamics (i.e., $\rho \approx 1$). This is achieved by testing the null hypothesis of no predictability:

$$\begin{aligned} H_0 : \quad & \beta = 0 \\ H_1 : \quad & \beta \neq 0 \end{aligned} \quad (4)$$

By extending the empirical likelihood (EL) framework of Cai et al. (2026), we propose a two-stage Orthogonalized Sieve-EL framework that achieves a standard χ_1^2 limiting distribution under the null.

2 Empirical Interpretation: Weather Shocks and Cocoa Futures

The framework can be applied to agricultural commodity markets by interpreting W_{t-1} as one or more local weather variables, such as precipitation, temperature, or humidity, and Y_t as excess returns on cocoa futures. The 2023–2024 cocoa market provides a useful example: severe weather conditions in West Africa

contributed to supply deficits and large price movements, while lagged prices and related financial predictors remained highly persistent.

This empirical setting fits the general framework for three reasons. First, weather variables are plausibly exogenous to contemporaneous financial-market innovations at the daily sampling frequency, while still being allowed to affect subsequent returns through the structural component $m(W_{t-1})$. Second, the relation between weather and supply is nonlinear, so a flexible function $m(W_{t-1})$ is more appropriate than a constant intercept or a linear control. Third, financial predictors such as lagged prices can be highly persistent, so inference on β must remain valid across near-unit-root persistence regimes.

Thus, in the cocoa application, the generic nuisance component $m(W_{t-1})$ has the interpretation of a nonlinear weather-response surface. The econometric method itself, however, does not depend on weather specifically; it only requires a stationary exogenous covariate vector satisfying the orthogonality and regularity conditions stated below.

3 The Orthogonalized Sieve-EL Framework

We handle the unknown nonparametric nuisance component $m(W_{t-1})$ by approximating it using a sieve basis of dimension K (e.g., polynomials or splines). Let $\mathbf{P}_K(W_{t-1}) = (p_1(W_{t-1}), \dots, p_K(W_{t-1}))'$ be the $K \times 1$ vector of these basis functions; when $d > 1$, the p_j are multivariate basis functions.

3.1 The Step-by-Step Algorithm

The estimation and inference procedure is implemented sequentially via the following three steps. The key idea is to profile out the flexible intercept component and then construct an EL moment that is orthogonal to the same sieve space.

Step 1: Profiling out the Nonparametric Nuisance Component

For each candidate parameter β under the null hypothesis, we profile out the unknown function $m(W_{t-1})$, which plays the role of a flexible intercept or exogenous-control component. Let $\mathbf{Y} = (Y_1, \dots, Y_T)'$ and $\mathbf{X}_{lag} = (X_0, \dots, X_{T-1})'$. Let $\mathbf{P}$ be the $T \times K$ matrix whose t -th row is $\mathbf{P}_K(W_{t-1})'$. The profile sieve coefficients for the nuisance component are:

$$\hat{\mathbf{c}}_K(\beta) = (\mathbf{P}'\mathbf{P})^{-1}\mathbf{P}'(\mathbf{Y} - \beta\mathbf{X}_{lag}) \quad (5)$$

The profile residual is defined using the sieve annihilator matrix $\mathbf{M}_K = \mathbf{I}_T - \mathbf{P}(\mathbf{P}'\mathbf{P})^{-1}\mathbf{P}'$:

$$\hat{\mathbf{u}}(\beta) = \mathbf{M}_K(\mathbf{Y} - \beta\mathbf{X}_{lag}) \quad (6)$$

Under the null hypothesis $\beta = \beta_0$, substituting the true DGP $\mathbf{Y} = \mathbf{P}\mathbf{c} + \mathbf{r} + \beta_0\mathbf{X}_{lag} + \mathbf{U}$ (where $\mathbf{r}$ is the approximation error) yields:

$$\hat{\mathbf{u}}(\beta_0) = \mathbf{M}_K(\mathbf{P}\mathbf{c} + \mathbf{r} + \mathbf{U}) = \mathbf{M}_K(\mathbf{r} + \mathbf{U}) \quad (7)$$

Step 2: The Sieve Weight Projection

In standard Empirical Likelihood, estimating a high-dimensional nuisance component can affect the limiting distribution. To neutralize this generated-nuisance effect, we construct an orthogonalized weight function. Let $\mathbf{w} = (w_1, \dots, w_T)'$ where $w_t = w(X_{t-1})$ is the raw weight (e.g., $\tanh(X_{t-1}/c_w)$). We project the weights onto the same sieve basis and define the orthogonalized weight $\mathbf{w}^c$ as the residual:

$$\mathbf{w}^c = \mathbf{M}_K\mathbf{w} \quad (8)$$

Let $w_t^c = (\mathbf{w}^c)_t$. By the properties of orthogonal projection, the orthogonalized weights are exactly orthogonal to the sieve basis in-sample:

$$\sum_{t=1}^T \mathbf{P}_K(W_{t-1})w_t^c \equiv \mathbf{0} \quad (9)$$

Step 3: Algebraic Elimination in the EL Moment Condition

For testing $H_0 : \beta = \beta_0$, the EL moment condition is evaluated as:

$$\widehat{Z}_t(\beta_0) = \widehat{u}_t(\beta_0)w_t^c \quad (10)$$

Summing the moment conditions across the sample to evaluate the numerator of the EL statistic:

$$\sum_{t=1}^T \widehat{Z}_t(\beta_0) = \widehat{\mathbf{u}}(\beta_0)' \mathbf{w}^c = (\mathbf{r} + \mathbf{U})' \mathbf{M}_K \mathbf{M}_K \mathbf{w} = (\mathbf{r} + \mathbf{U})' \mathbf{w}^c \quad (11)$$

The fitted sieve component is thereby eliminated algebraically. The remaining approximation residual is controlled by the sieve rate conditions, leaving:

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T \widehat{Z}_t(\beta_0) = \frac{1}{\sqrt{T}} \sum_{t=1}^T U_t w_t^c + \underbrace{\frac{1}{\sqrt{T}} \sum_{t=1}^T r_t w_t^c}_{= o_p(1)} \quad (12)$$

Because the moment condition is orthogonal to the fitted nuisance space, the sample variance of $\widehat{Z}_t(\beta_0)$ consistently estimates the relevant innovation variance under the stated rate conditions. As a result, the Empirical Likelihood ratio statistic (where $\widehat{\lambda}$ is the Lagrange multiplier) achieves the standard chi-squared limit:

$$\ell_T(\beta_0) = 2 \sum_{t=1}^T \log(1 + \widehat{\lambda} \widehat{Z}_t(\beta_0)) \xrightarrow{d} \chi_1^2 \quad (13)$$

3.2 Comparison with Cai et al. (2026)

Our framework differs from the approach of Cai et al. (2026) in two fundamental aspects:

- **Nuisance Dimensionality:** Cai et al. (2026) handle a fixed scalar intercept α , whereas our model contains an infinite-dimensional nuisance function $m(W_{t-1})$, which requires a sieve approximation of dimension $K \rightarrow \infty$.
- **Orthogonalization Space:** The true generalization is that the demeaning against a constant 1 is replaced by residualizing against the sieve basis $\mathbf{P}_K(W)$. In Cai et al. (2026), the weight centering $w^c(x) = w(x) - \frac{1}{T} \sum w(X_s)$ represents a projection onto the constant subspace $\text{span}\{\mathbf{1}\}$. In contrast, our orthogonalized weight $\mathbf{w}^c$ is projected onto the K -dimensional sieve space $\text{span}\{\mathbf{P}_K(W_{t-1})\}$. We keep the AR(1) equation as the persistence DGP, but we do not use AR(1) OLS residuals in the EL statistic.

4 The Growth Rate of the Sieve Dimension K

To establish the asymptotic validity of our test statistic, we must place bounds on the growth rate of the tuning parameter K relative to the sample size T .

4.1 Lower Bound (Bias Elimination)

Assume the true nuisance function $m(W)$ has smoothness s and that W is d -dimensional. Sieve theory implies that the approximation error r_t satisfies $\sup |r_t| = O_p(K^{-s/d})$. To ensure that the approximation bias does not distort the asymptotic distribution, we require $T^{-1/2} \sum_{t=1}^T r_t w_t^c = o_p(1)$, which translates to:

$$\sqrt{T} K^{-s/d} \rightarrow 0 \quad \implies \quad K \gg T^{\frac{d}{2s}} \quad (14)$$

4.2 Upper Bound (Variance Control and Leakage Mitigation)

Estimating K nuisance parameters introduces in-sample variance inflation and potential projection leakage. Under mixing conditions for the exogenous covariate process W_t , this leakage does not accumulate, and the leading variance penalty scales as $K/\sqrt{T}$. To prevent variance inflation from affecting the EL ratio, we require:

$$\frac{K}{\sqrt{T}} \rightarrow 0 \quad \implies \quad K \ll T^{\frac{1}{2}} \quad (15)$$

4.3 Growth rate of K

Combining these bounds yields the formal asymptotic growth condition:

$$T^{\frac{d}{2s}} \ll K \ll T^{\frac{1}{2}} \quad (16)$$

This rate window is nonempty when $s > d$.